

EGFR fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes EGFR gene fusions from the msk_impact_50k_2026 study, covering 55 fusion events. 28/55 fusions (50.9%) were in-frame. 41/55 fusions (74.5%) retained the Protein tyrosine and serine/threonine kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0114539$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for EGFR.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 55 analyzed fusion events, identifying 31 distinct fusion partner genes. The most frequent partner was SEPTIN14, observed in 18 events.

Domain retention

Protein tyrosine and serine/threonine kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 25/28 (89.3%) of in-frame fusions retained the domain, $p=0.0114539$ (statistically significant at $\alpha=0.05$). The Protein tyrosine and serine/threonine kinase appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.247525.

Domain disruption

Domain-disruption analysis was skipped: EGFR has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 55 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 982 aa (permutation-corrected $p=0.128713$, not statistically significant at $\alpha=0.05$). This is 17 aa from the nearest configured domain boundary at 965 aa.

Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: EGFR has no group_field configured; confidence-stats analysis was skipped.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

Joint partner

Joint partner reported: EGFR has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SEPTIN14	18	0.327272727272727	0.060638136511060495	None	0.9786998132586368	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.3716376061405412	1
RAD51	2	0.036363636363636	0.060638136511060495	None	0.9989495798319328	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2513015171594942	2
LANCL2	3	0.054545454545454	0.060638136511060495	None	0.9618347338935574	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.25114054367919303	3
ELAPOR2	1	0.018181818181818	0.060638136511060495	None	0.9989495798319328	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2435093093672864	4
PKD1L1	1	0.018181818181818	0.060638136511060495	None	0.9989495798319328	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2435093093672864	5
VPS41	1	0.018181818181818	0.060638136511060495	None	0.9957983193277311	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2428340392592432	6
ZNF713	1	0.018181818181818	0.060638136511060495	None	0.9401260504201681	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2309042673504797	7
VWC2	2	0.036363636363636	0.060638136511060495	None	0.9028361344537815	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2307057788641761	8
NIPSNAP2	3	0.054545454545454	0.060638136511060495	None	0.8658963585434174	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2305823203898773	9
TNRC18	1	0.018181818181818	0.060638136511060495	None	0.9233193277310925	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2273028267742492	10
BLTP3B	1	0.018181818181818	0.060638136511060495	None	0.9023109243697479	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22280102605396107	11
CDH7	1	0.018181818181818	0.060638136511060495	None	0.832983193277311	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2079450836770103	12
FADD	2	0.036363636363636	0.060638136511060495	None	0.6901260504201681	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.18512504657125892	13
EEA1	1	0.018181818181818	0.060638136511060495	None	0.6911764705882353	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17755792881506555	14

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIF5B	1	0.01818181818181818	0.060638136511060495	None	0.6911764705882353	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17755792881506555	15
VSTM2A	1	0.01818181818181818	0.060638136511060495	None	0.6911764705882353	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17755792881506555	16
CADPS	1	0.01818181818181818	0.060638136511060495	None	0.6901260504201681	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17733283877905112	17
NUMA1	1	0.01818181818181818	0.060638136511060495	None	0.6901260504201681	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17733283877905112	18
PDE1C	1	0.01818181818181818	0.060638136511060495	None	0.6523109243697479	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16922959748253252	19
YIF1B	1	0.01818181818181818	0.060638136511060495	None	0.6407563025210083	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16675360708637407	20
TUT7	1	0.01818181818181818	0.060638136511060495	None	0.6271008403361344	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1638274366181868	21
CSF2RA	1	0.01818181818181818	0.060638136511060495	None	0.3214285714285714	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.09832623613799471	22
DDC	1	0.01818181818181818	0.060638136511060495	None	0.2804621848739496	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08954772473343288	23
GARS1	1	0.01818181818181818	0.060638136511060495	None	0.2804621848739496	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08954772473343288	24
SEL1L	1	0.01818181818181818	0.060638136511060495	None	0.2804621848739496	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08954772473343288	25
TNS3	1	0.01818181818181818	0.060638136511060495	None	0.2804621848739496	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08954772473343288	26
PCDH15	1	0.01818181818181818	0.060638136511060495	None	0.2710084033613446	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08752191440930324	27
VOPP1	1	0.01818181818181818	0.060638136511060495	None	0.23004201680672265	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07874340300474139	28
SCAF4	1	0.01818181818181818	0.060638136511060495	None	0.11764705882352944	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.05465876915119999	29
ZBPB	1	0.01818181818181818	0.060638136511060495	None	0.006302521008403339	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.030799225333672974	30

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PL0D3	1	0.01818181818181818	0.060638136511060495	None	0.0	None	['cutpoint_proximity', 'domain_retention', 'recurrence']	0.029448685117586543	31

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
30	1	40	0	14	None	1.0	-0.0
36	2	39	0	14	None	1.0	-0.0
142	3	38	0	14	None	0.5624166190203925	0.24994185469837954
249	3	38	1	13	1.0263157894736843	1.0	-0.0
288	3	38	2	12	0.47368421052631576	0.5924468510484049	0.22735060452753406
297	7	34	2	12	1.2352941176470589	1.0	-0.0
336	8	33	2	12	1.4545454545454546	1.0	-0.0
599	9	32	2	12	1.6875	0.7093311767774617	0.14915095146564436
627	9	32	3	11	1.03125	1.0	-0.0
640	10	31	3	11	1.1827956989247312	1.0	-0.0
651	11	30	3	11	1.3444444444444446	1.0	-0.0
687	13	28	5	9	0.8357142857142857	1.0	-0.0
688	15	26	6	8	0.7692307692307693	0.7548366596998944	0.12214701587495107
823	15	26	7	7	0.5769230769230769	0.528760430668828	0.27674105235972185
875	15	26	9	5	0.32051282051282054	0.1175979559898796	0.9296002268145651
981	15	26	10	4	0.23076923076923078	0.031971018585759146	1.4952435270336717
982	21	20	12	2	0.175	0.02895546314423952	1.5382694841107174
983	31	10	13	1	0.23846153846153847	0.25480492869908067	0.5937921756847577
986	32	9	13	1	0.27350427350427353	0.4229368950729017	0.37372442734387856
994	33	8	13	1	0.3173076923076923	0.4208384837822578	0.3758845527657329
1003	34	7	13	1	0.37362637362637363	0.663943317750199	0.17786899569082928
1039	35	6	13	1	0.44871794871794873	0.663943317750199	0.17786899569082928
1054	37	4	13	1	0.7115384615384616	1.0	-0.0
1055	38	3	14	0	0.0	0.5624166190203926	0.24994185469837946
1060	39	2	14	0	0.0	1.0	-0.0
1075	40	1	14	0	0.0	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

25	16
3	11

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	55	41	3	11	0.10304912478825523

frequency tables

Partner_gene_counts

Partner_gene	Event_count
BLTP3B	1
CADPS	1
CDH7	1
CSF2RA	1
DDC	1
EEA1	1
ELAPOR2	1
FADD	2
GARS1	1
KIF5B	1
LANCL2	3
NIPSNAP2	3
NUMA1	1
PCDH15	1
PDE1C	1
PKD1L1	1
PLOD3	1
RAD51	2
SCAF4	1
SEL1L	1
SEPTIN14	18

Partner_gene	Event_count
TNRC18	1
TNS3	1
TUT7	1
VOPP1	1
VPS41	1
VSTM2A	1
VWC2	2
YIF1B	1
ZNF713	1
ZPBP	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 1404-T01-I M6-2	P-0031 404-T0 1-IM6		BLTP 3B-E GFR	BLT P3B	unknown	five_prime	552702 72	27	1075	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - UHRF1BP1L (NM_015054) Rearrangement :t(7;12)(p11.2;q23.1)(chr7:g.55270272::chr12:g.100460508) Protein Fusion: mid-exon {EGFR:UHRF1BP1L}	NA
EVT-P-005 0679-T02-I M6-3	P-0050 679-T0 2-IM6		CADP S-EG FR	CA DPS	out-of-frame	three_prime	552410 99	17	687	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			CADPS (NM_003716) - EGFR (NM_005228) fusion: t(3;7)(p14.2;p11.2)(chr3:g.62537410::chr7:g.55241099) Protein Fusion: out of frame {CADPS:EGFR}	NA
EVT-P-003 8683-T01-I M6-4	P-0038 683-T0 1-IM6		CSF2 RA-E GFR	CSF 2RA	unknown	three_prime	552242 20	9	336	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			CSF2RA (NM_001161530) - EGFR (NM_005228) rearrangement: t(7;X)(p11.2;p22.33)(chr7:g.55224220::chrX:g.1401662) Protein Fusion: mid-exon {CSF2RA:EGFR}	NA
EVT-P-001 6795-T01-I M6-5	P-0016 795-T0 1-IM6		DDC- EGFR	DD C	out-of-frame	three_prime	552222 04	7	297	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			DDC (NM_000790) - EGFR (NM_005228) Rearrangement : c.435+4059 :DDC_c.889+359:EGFRinv Protein Fusion: out of frame {DDC:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 8389-T02-I M6-6	P-0018 389-T0 2-IM6		EEA1-EGFR	EEA1	in-frame	threonine_prime	55241575	18	688	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EEA1 (NM_003566) - EGFR (NM_005228) rearrangement: t(7;12)(p11.2;q22)(chr7:g.55241575::chr12:g.932211089) Protein Fusion: in frame {EEA1:EGFR}	NA
EVT-P-006 4660-T01-I M7-9	P-0064 660-T0 1-IM7		EGFR-CDH7	CDH7	out-of-frame	five_prime	55254206	20	823	True	disrupted	0.43873517786561267	True			Protein tyrosine and serine/threonine kinase	EGFR (NM_005228) - CDH7 (NM_033646) rearrangement: t(7;18)(p11.2;q22.1)(chr7:g.55254206::chr18:g.63486938) Protein Fusion: out of frame {EGFR:CDH7}	NA
EVT-P-003 4892-T01-I M6-497	P-0034 892-T0 1-IM6		EGFR-ELAPOR2	ELAPOR2	unknown	threonine_prime	55268103	24	981	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) rearrangement: c.2943:EGFR_chr7:g.86587163del Antisense Fusion	NA
EVT-P-005 9222-T01-I M7-500	P-0059 222-T0 3-IM7		EGFR-LANCL2	LANCL2	in-frame	five_prime	55259704	21	875	True	disrupted	0.6442687747035574	True			Protein tyrosine and serine/threonine kinase	EGFR (NM_005228) - LANCL2 (NM_018697) rearrangement: c.2625+137:EGFR_c.1008+3812:LANCL2del Protein Fusion: in frame {EGFR:LANCL2}	NA
EVT-P-005 7880-T01-I M6-501	P-0057 880-T0 1-IM6		EGFR-LANCL2	LANCL2	in-frame	five_prime	55268871	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - LANCL2 (NM_018697) fusion: c.2947-10:EGFR_c.205-8690:LANCL2del Protein Fusion: in frame {EGFR:LANCL2}	NA
EVT-P-006 3142-T01-I M7-503	P-0063 142-T0 1-IM7		EGFR-PCDH15	PCDH15	unknown	five_prime	55221820	7	288	False	lost	0.0	False		Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - PCDH15 (NM_001142763) rearrangement: t(7;10)(p11.2;q21.1)(chr7:g.55221820::chr10:g.56187463) Protein Fusion: mid-exon {EGFR:PCDH15}	NA
EVT-P-004 2236-T01-I M6-504	P-0042 236-T0 1-IM6		EGFR-PKD1L1	PKD1L1	in-frame	five_prime	55268824	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - PKD1L1 (NM_138295) fusion: c.2947-57:EGFR_c.8527-7825:PKD1L1inv Protein Fusion: in frame {EGFR:PKD1L1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 1149-T04-I M6-505	P-0041 149-T0 4-IM6		EGFR -RAD 51	RAD D51	in-frame	five_prime	552687 99	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.55268799::chr15:g.40996189) Protein Fusion: in frame {EGFR:RAD51}	NA
EVT-P-004 1149-T01-I M6-506	P-0041 149-T0 1-IM6		EGFR -RAD 51	RAD D51	in-frame	five_prime	552687 99	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - RAD51 (NM_002875) fusion: t(7;15)(p11.2;q15.1)(chr7:g.55268799::chr15:g.40996189) Protein Fusion: in frame {EGFR:RAD51}	NA
EVT-P-003 4594-T01-I M6-507	P-0034 594-T0 1-IM6		EGFR -SEP TIN14	SEP TIN 14	in-frame	five_prime	552685 44	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-110:EGFR_c.1120-1204:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-003 8138-T01-I M6-508	P-0038 138-T0 1-IM6		EGFR -SEP TIN14	SEP TIN 14	in-frame	five_prime	552683 81	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+275:EGFR_c.1119+2043:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-003 7526-T01-I M6-509	P-0037 526-T0 1-IM6		EGFR -SEP TIN14	SEP TIN 14	unknown	five_prime	552689 16	25	994	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2982:EGFR_c.1119+1151:SEPT14inv Protein Fusion: mid-exon {EGFR:SEPT14}	NA
EVT-P-003 9255-T02-I M6-510	P-0039 255-T0 2-IM6		EGFR -SEP TIN14	SEP TIN 14	in-frame	five_prime	552708 07	27	1091	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3271+489:EGFR_c.818-3507:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-003 2604-T01-I M6-512	P-0032 604-T0 1-IM6		EGFR -SEP TIN14	SEP TIN 14	in-frame	five_prime	552687 71	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-110:EGFR_c.1120-1204:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 1753-T01-I M6-513	P-0031 753-T0 1-IM6		EGFR -SEPTIN14	SEPTIN14	in-frame	five_prime	55268412	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2946+306:EGFR_c.1119+2531:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-000 5701-T01-I M5-514	P-0005 701-T0 1-IM5		EGFR -SEPTIN14	SEPTIN14	in-frame	five_prime	55269490	26	1054	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement : c.3162+15:EGFR_c.1119+2833:SEPT14inv Protein fusion: in frame {EGFR:SEPT14}	NA
EVT-P-004 3525-T01-I M6-515	P-0043 525-T0 1-IM6		EGFR -SEPTIN14	SEPTIN14	in-frame	five_prime	55268859	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2947-22:EGFR_c.1120-807:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-002 9159-T01-I M6-516	P-0029 159-T0 1-IM6		EGFR -SEPTIN14	SEPTIN14	in-frame	three_prime	55269850	27	1055	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		SEPT14 (NM_207366) - EGFR (NM_005228) rearrangement: c.1119+1369:SEPT14_c.3163-360:EGFRinv Protein Fusion: in frame {SEPT14:EGFR}	NA
EVT-P-002 8241-T01-I M6-517	P-0028 241-T0 1-IM6		EGFR -SEPTIN14	SEPTIN14	in-frame	five_prime	55268525	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-24 fused to SEPT14 exon 10): c.2947-356:EGFR_c.1119+4312:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-002 5191-T02-I M6-518	P-0025 191-T0 2-IM6		EGFR -SEPTIN14	SEPTIN14	in-frame	five_prime	55269542	26	1054	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.3162+67:EGFR_c.1120-3396:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-006 1245-T01-I M7-519	P-0061 245-T0 1-IM7		EGFR -SEPTIN14	SEPTIN14	in-frame	five_prime	55268815	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) rearrangement: c.2947-66:EGFR_c.1119+2973:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 4220-T02-I M6-520	P-0014 220-T0 2-IM6		EGFR -SEPTIN14	SEPTIN14	unknown	five_prime	55268942	25	1003	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion (EGFR exons 1-25 fused with SEPT14 exon 10): c.3008:EGFR_c.1120-4102:SEPT14inv Protein Fusion: mid-exon {EGFR:SEPT14}	NA
EVT-P-002 3043-T01-I M6-521	P-0023 043-T0 1-IM6		EGFR -SEPTIN14	SEPTIN14	in-frame	five_prime	55268179	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) Fusion (EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+73:EGFR_c.1120-180:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-001 9603-T01-I M6-522	P-0019 603-T0 1-IM6		EGFR -SEPTIN14	SEPTIN14	in-frame	five_prime	55268171	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) Fusion ((EGFR exons 1-24 fused with SEPT14 exon 10) : c.2946+65:EGFR_c.1119+3306:SEPT14 Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-004 7356-T01-I M6-523	P-0047 356-T0 1-IM6		EGFR -SEPTIN14	SEPTIN14	in-frame	five_prime	55268316	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion: c.2946+210:EGFR_c.1120-2817:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-001 6256-T01-I M6-524	P-0016 256-T0 1-IM6		EGFR -SEPTIN14	SEPTIN14	in-frame	five_prime	55268390	24	982	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - SEPT14 (NM_207366) fusion(EGFR exons 1-24 fused to SEPT14 exons 9-10): c.2946+284:EGFR_c.1120-3940:SEPT14inv Protein Fusion: in frame {EGFR:SEPT14}	NA
EVT-P-001 5179-T01-I M6-525	P-0015 179-T0 1-IM6		EGFR -SEPTIN14	SEPTIN14	in-frame	five_prime	55268650	25	983	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005229) - SEPT14 (NM_014857) fusion (EGFR exons 1-24 fused with SEPT14 exon 10): c.2947-231:EGFR_c.1119+3910:SEPT14inv Protein fusion: in frame (EGFR-SEPT14)	NA
EVT-P-005 8375-T01-I M6-526	P-0058 375-T0 1-IM6		EGFR -TNRC18	TNRC18	out-of-frame	five_prime	55269897	27	1055	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - TNRC18 (NM_001080495) fusion: c.3163-313:EGFR_c.5720-6144:TNRC18inv Protein Fusion: out of frame {EGFR:TNRC18}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-001 6009-T01-I M6-527	P-0016 009-T0 1-IM6		EGFR -VOP P1	VO PP1	in-frame	five_prime	552203 93	6	249	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - VOPP1 (NM_030796): c.747+36:EGFR_c.55-23161VOPP1inv Protein Fusion: in frame {EGFR:VOPP1}	NA
EVT-P-003 9262-T01-I M6-529	P-0039 262-T0 1-IM6		EGFR -VST M2A	VST M2A	out-of-frame	five_prime	552415 09	18	688	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - VSTM2A (NM_182546) rearrangement: c.2062-105:EGFR_c.634+2190:VSTM2A dup Protein Fusion: out of frame {EGFR:VSTM2A}	NA
EVT-P-004 2236-T01-I M6-530	P-0042 236-T0 1-IM6		EGFR -VWC 2	VW C2	unknown	five_prime	552702 25	27	1060	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - VWC2 (NM_198570) fusion: c.3178:EGFR_c.827-13914:VWC2dup Protein Fusion: mid-exon {EGFR:VWC2}	NA
EVT-P-004 3097-T01-I M6-531	P-0043 097-T0 1-IM6		EGFR -ZNF7 13	ZNF 713	in-frame	five_prime	552694 00	26	1039	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) - ZNF713 (NM_182633) fusion: c.3115-28:EGFR_c.49-962:ZNF713del Protein Fusion: in frame {EGFR:ZNF713}	NA
EVT-P-002 7944-T01-I M6-534	P-0027 944-T0 1-IM6		FADD -EGF R	FAD D	unknown	five_prime	552410 77	17	687	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330) Protein Fusion: mid-exon {EGFR:FADD}	NA
EVT-P-002 7944-T03-I M6-535	P-0027 944-T0 3-IM6		FADD -EGF R	FAD D	unknown	five_prime	552410 77	17	687	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - FADD (NM_003824) rearrangement: t(7;11)(p11.2;q13.3)(chr7:g55241077::chr11:g70052330) Protein Fusion: mid-exon {EGFR:FADD}	NA

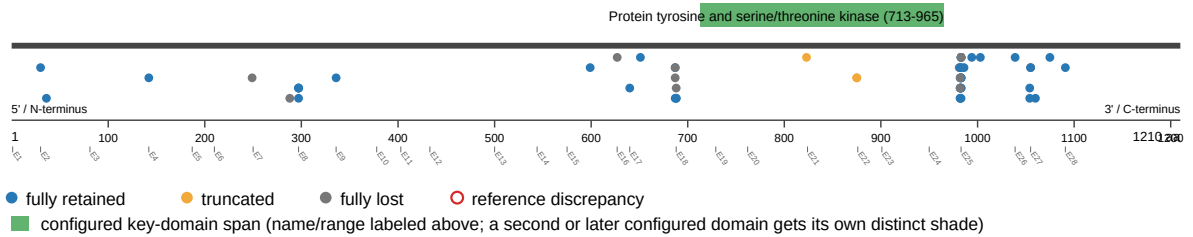
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-006 3743-T01-I M7-536	P-0063 743-T0 1-IM7		GARS 1-EGFR	GARS1	out-of-frame	three_prime	55222368	7	297	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			GARS (NM_002047) - EGFR (NM_005228) fusion: c.1032-338:GARS_c.889+523:EGFRdel Protein Fusion: out of frame {GARS:EGFR}	NA
EVT-P-002 1038-T01-I M6-538	P-0021 038-T0 1-IM6		KIF5B -EGFR	KIF5B	in-frame	three_prime	55241486	18	688	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			KIF5B (NM_004521) - EGFR (NM_005228) rearrangement: t(7;10)(p11.2;q11.22)(chr7:g.55241486::chr10:g.32312089) Protein Fusion: in frame {KIF5B:EGFR}	NA
EVT-P-002 5018-T01-I M6-539	P-0025 018-T0 1-IM6		LANC L2-EGFR	LANCL2	out-of-frame	three_prime	55268537	25	983	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		LANCL2 (NM_018697) - EGFR (NM_005228) rearrangement: c.1259-255:LANCL2_c.2947-344:EGFRdup Protein Fusion: out of frame {LANCL2:EGFR}	NA
EVT-P-005 9222-T01-I M7-540	P-0059 222-T0 1-IM7		NIP SNA P2 -EGFR	NIP SNA P2	unknown	three_prime	55233046	15	599	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			GBAS (NM_001483) - EGFR (NM_005228) fusion: c.92+717:GBAS_c.1796:EGFRdup Protein Fusion: mid-exon {GBAS:EGFR}	NA
EVT-P-003 7759-T01-I M6-541	P-0037 759-T0 1-IM6		NIP SNA P2 -EGFR	NIP SNA P2	out-of-frame	three_prime	55268444	24	982	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		GBAS (NM_001483) - EGFR (NM_005228) Rearrangement: c.2946+338:EGFR_c.279-1159:GBASdup Protein Fusion: out of frame {GBAS:EGFR}	NA
EVT-P-003 7759-T03-I M6-542	P-0037 759-T0 3-IM6		NIP SNA P2 -EGFR	NIP SNA P2	out-of-frame	three_prime	55268444	24	982	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		GBAS (NM_001483) - EGFR (NM_005228) rearrangement: c.279-1159:GBAS_c.2946+338:EGFRdup Protein Fusion: out of frame {GBAS:EGFR}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 6028-T01-I M6-543	P-0036 028-T0 1-IM6		NUMA1-EGFR	NUMA1	in-frame	threonine_prime	55241128	17	687	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NUMA1 (NM_006185) - EGFR (NM_005228) rearrangement: t(7;11)(p11.2;q13.4)(chr7:g.55241128::chr11:g.71728415) Protein Fusion: in frame {NUMA1:EGFR}	NA
EVT-P-006 5470-T01-I M7-545	P-0065 470-T0 1-IM7		PDE1C-EGFR	PDE1C	unknown	threonine_prime	55240707	17	651	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			PDE1C (NM_001191058) - EGFR (NM_005228) fusion: c.86-15224:PDE1C_c.1951:EGFRinv Protein Fusion: mid-exon {PDE1C:EGFR}	NA
EVT-P-005 9032-T01-I M7-546	P-0059 032-T0 1-IM7		PLOD3-EGFR	PLOD3	out-of-frame	threonine_prime	55179084	2	30	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			PLOD3 (NM_001084) - EGFR (NM_005228) fusion: c.2062-9:PLOD3_c.89-30895:EGFRinv Protein Fusion: out of frame {PLOD3:EGFR}	NA
EVT-P-002 8484-T02-I M6-548	P-0028 484-T0 2-IM6		SCAF4-EGFR	SCAF4	out-of-frame	threonine_prime	55214145	4	142	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			SCAF4 (NM_020706) - EGFR (NM_005228) fusion: t(7;21)(p11.2;q22.11)(chr7:g.55214145::chr21:g.33094864) Protein Fusion: out of frame {SCAF4:EGFR}	NA
EVT-P-003 0910-T01-I M6-551	P-0030 910-T0 1-IM6		SEL1L-EGFR	SEL1L	in-frame	threonine_prime	55222485	7	297	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			SEL1L (NM_005065) - EGFR (NM_005228) rearrangement: t(7;14)(p11.2;q31.1)(chr7:g.55222485::chr14:g.81986576) Protein Fusion: in frame {SEL1L:EGFR}	NA
EVT-P-005 2976-T02-I M7-552	P-0052 976-T0 2-IM7		TNS3-EGFR	TNS3	in-frame	threonine_prime	55222307	7	297	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			TNS3 (NM_022748) - EGFR (NM_005228) fusion: c.1025-2026:TNS3_c.889+462:EGFRinv Protein Fusion: in frame {TNS3:EGFR}	NA
EVT-P-000 6873-T01-I M5-554	P-0006 873-T0 1-IM5		TUT7-EGFR	TUT7	unknown	five_prime	55238031	16	627	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		EGFR (NM_005228) - ZCCHC6 (NM_024617) rearrangement : t(7;9)(chr7p11.2;chr9q21.33)(chr7:g.55238031::chr9:g.88930920) Protein fusion: mid-exon (EGFR-ZCCHC6)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015419-T01-IM6-555	P-0015419-T01-IM6		VPS41-EGFR	VPS41	unknown	five_prime	55268890	25	986	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228)- VPS41 (NM_014396) Rearrangement : c.2956:EGFR_c.247-4922:VPS41inv Protein fusion: mid-exon (EGFR-VPS41)	NA
EVT-P-0030222-T01-IM6-557	P-0030222-T01-IM6		VWC2-EGFR	VWC2	out-of-frame	three_prime	55259621	21	875	True	disrupted	0.35968379446640314	True			Protein tyrosine and serine/threonine kinase	VWC2 (NM_198570) - EGFR (NM_005228) rearrangement: c.827-37789:VWC2_c.2625+54:EGFRdel Protein Fusion: out of frame {VWC2:EGFR}	NA
EVT-P-0019922-T01-IM6-558	P-0019922-T01-IM6		YIF1B-EGFR	YIF1B	unknown	three_prime	55238939	16	640	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			EGFR (NM_005228) rearrangement: t(7;19)(p22.1;q13.2)(chr7:g.5523893::chr19:g.38806683) Antisense Fusion	NA
EVT-P-0030222-T01-IM6-559	P-0030222-T01-IM6		ZPBP-EGFR	ZPBP	unknown	three_prime	55209996	2	36	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase			ZPBP (NM_007009) - EGFR (NM_005228) rearrangement: c.487+5719:ZPBP_c.106:EGFRinv Protein Fusion: mid-exon {ZPBP:EGFR}	NA

domain retention outliers

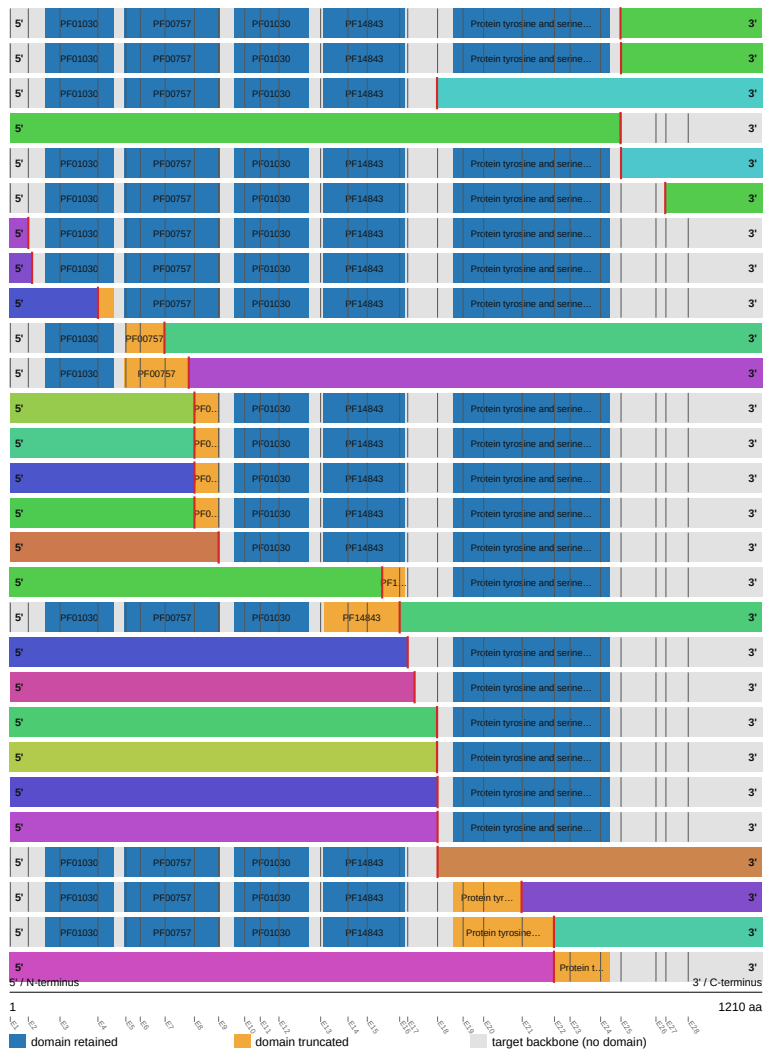
EGFR domain-retention track



fusion schematic

EGFR fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (EGFR portion is domain-colored, exon ticks)



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 41 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

SEPTIN14 (x6) – Protein tyrosine and serine/threonine kinase retained

SEPTIN14 (x6) – Protein tyrosine and serine/threonine kinase retained

FADD (x2) – Protein tyrosine and serine/threonine kinase lost

NIPSNAP2 (x2) – Protein tyrosine and serine/threonine kinase lost

RAD51 (x2) – Protein tyrosine and serine/threonine kinase retained

SEPTIN14 (x2) – Protein tyrosine and serine/threonine kinase retain

PI OD3 – Protein tyrosine and serine/threonine kinase retained

7PBP – Protein tyrosine and serine/threonine kinase retained

SCAF4 – Protein tyrosine and serine/threonine kinase retained

VOPR1 – Protein tyrosine and serine/threonine kinase lost

PCDU15 Protein tyrosine and serine/threonine kinase test

DDC Protein turning and aging/lysine kinase retained

045361 Protein binding and α - β transition in the α -subunit of the

1100 Protein Synthesis and Gene Expression Release Rates

OSI 2104—1-*osin* tyrosine and serine/threonine kinase retained

NIP-SNAF-2 = Protein tyrosine and serine/threonine kinase rec

1017 - Protein tyrosine and serine/threonine kinase lost

YIF1B – Protein tyrosine and serine/threonine kinase retained

PDE1C – Protein tyrosine and serine/threonine kinase retained

CADPS – Protein tyrosine and serine/threonine kinase retained

NUMA1 – Protein tyrosine and serine/threonine kinase retained

EEA1 – Protein tyrosine and serine/threonine kinase retained

KIF5B – Protein tyrosine and serine/threonine kinase retained

VSTM2A – Protein tyrosine and serine/threonine kinase lost

CDH7 – Protein tyrosine and serine/threonine kinase truncated

LANCL2 – Protein tyrosine and serine/threonine kinase truncated

WVC2 – Protein tyrosine and serine/threonine kinase truncated

■ breakpoint

intragenic deletion schematic

EGFR intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



- 6-exon deletion (in-frame) (x279)
- 3-exon deletion (out-of-frame) (x24)
- 3-exon deletion (out-of-frame) (x19)
- 2-exon deletion (in-frame) (x5)
- 2-exon deletion (in-frame) (x5)
- 1-exon deletion (out-of-frame) (x5)
- 2-exon deletion (in-frame) (x3)
- 1-exon deletion (out-of-frame) (x3)
- 3-exon deletion (in-frame) (x2)
- 6-exon deletion (out-of-frame) (x2)
- 4-exon deletion (in-frame)
- 2-exon deletion (in-frame)
- 5-exon deletion (in-frame)
- 3-exon deletion (in-frame)
- 3-exon deletion (out-of-frame)
- 2-exon deletion (in-frame)
- 2-exon deletion (in-frame)

reference comparison

Reference vs msk_impact_50k_2026

In-frame



reference 0.0%

run 50.9%

Domain retained



reference 0.0%

run 74.5%

reference current run